Sarp Aygen

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Education

Queen's University | Kingston, ON
Bachelor of Computing, Software Design | Sep 2024 Expected graduation: 2028

Relevant coursework: Computer Systems, System Level Programming, Data Structures and Algorithms, Programming Paradigms, Discrete Structures, Computer Architecture

Technical Skills

Languages: Python, C, C++, Java, Bash, SQL

Testing: Functional, regression, and negative testing, test case design, test automation, defect reproduction, root cause analysis, diagnostic logging

Frameworks and data: OpenCV, NumPy, Tesseract OCR, PySide6, Spring Boot, PostgreSQL

Systems and tools: Windows, Linux/Unix, Windows API, Git, GitHub, PC hardware troubleshooting

Foundations: OOP, multithreading, REST APIs, familiarity with LLM and RAG concepts

Projects

Computer Vision Desktop Automation | GitHub

Python, OpenCV, NumPy, Tesseract OCR, PySide6, Windows API

- Built a Windows application using OpenCV template matching and Tesseract OCR to detect on screen events and coordinate input across up to four application windows.
- Observed approximately 94% detection accuracy across about 100 manual in game tests.
- Wrote functional, regression, and negative tests using screenshots and mocked Windows input, covering event transitions, stale OCR results, disconnections, and dropped or delayed inputs.
- Reproduced and diagnosed detection and timing defects using screenshots and logs, implemented bounded retries, independent window recovery, and visual checks before marking recovery complete.
- Reduced repeated processing by caching image templates and batching PySide6 log updates, separated time sensitive monitoring from OCR and background tasks.

SpendWise Expense Tracking Software | GitHub

Java, Spring Boot, PostgreSQL | In development

- Built a Java/Spring Boot REST API with PostgreSQL persistence to create, retrieve, and categorize financial transactions.
- Separated request handling, business logic, and data access into controller, service, and repository layers.
- Implemented request validation and centralized exception handling to return field specific errors and appropriate HTTP status codes.

Custom PC Builds and Hardware Troubleshooting

Personal projects

- Assembled and configured custom PCs for personal use and others, selecting components for compatibility, cooling, power requirements, and budget.
- Diagnosed GPU/RAM detection, display, and PCIe issues using BIOS inspection, CMOS resets, and systematic hardware troubleshooting.